

Problem Statement 5:

Quantum-Inspired Cyber Threat Detection for Digital Signature Security

Background:

The rapid advancement of quantum computing poses a serious threat to classical public-key cryptographic systems such as RSA and Elliptic Curve Cryptography (ECC), which can be broken by algorithms like Shor's algorithm. This vulnerability endangers the security of critical digital infrastructures. Quantum Digital Signature (QDS) protocols offer information-theoretic security by exploiting fundamental principles of quantum mechanics. Among these, teleportation-based QDS protocols are particularly promising because they enable secure signature generation and verification through quantum teleportation and entanglement, while reducing some of the practical deployment complexities associated with earlier QDS schemes.

Description

This problem focuses on developing a quantum-inspired cyber threat detection framework specifically designed for Quantum Digital Signature (QDS) systems. The framework will detect threats to the integrity and authenticity of digital signatures—such as forgery, impersonation, replay attacks, and quantum channel manipulation—without relying on artificial intelligence or machine learning techniques. Instead, it will utilize quantum principles including Pauli eigenstates, projective measurements, and statistical analysis of measurement outcomes to evaluate forgery probabilities and verification accuracy, while preserving information-theoretic security guarantees.

Objectives

- Design a quantum-inspired threat detection framework for teleportation-based Quantum Digital Signature protocols.
- Detect digital signature forgery, impersonation, replay attacks, and unauthorized verification attempts.
- Utilize Pauli eigenstates, quantum measurement analysis, and statistical threshold methods for threat identification.
- Ensure efficient verification algorithms that maintain information-theoretic security.

- Evaluate the framework through forgery probability analysis, attack simulations, and performance metrics.

Expected Solution:

A software framework for Quantum-Inspired Cyber Threat Detection tailored to teleportation-based Quantum Digital Signature protocols. The solution will simulate quantum public key distribution using Bell-state entanglement and quantum teleportation, apply Pauli correction operations and projective measurements for signature verification, and detect malicious activities through statistical evaluation and threshold-based decision rules. The framework will include mathematical modelling, attack simulation capabilities, security analysis, and performance evaluation, ensuring deterministic acceptance of legitimate signatures, low computational complexity, and strong security guarantees.

Delivery Table (Expected Deliverables)

S.No	Deliverable	Description	Key Components / Metrics
1	Mathematical Model of Teleportation-based QDS	Formal description of the signature protocol	Bell-state entanglement, Quantum teleportation process, Pauli correction operations, Projective measurement rules
2	Quantum-Inspired Threat Detection Framework	Core detection engine	Detection of forgery, impersonation, replay attacks, and channel manipulation using Pauli eigenstates and measurement statistics
3	Signature Generation & Verification Module	Implementation of QDS operations	Quantum public key distribution simulation, Signature generation, Verification algorithm with Pauli corrections
4	Attack Simulation Module	Controlled simulation of cyber threats	Forgery attempts, Impersonation attacks, Replay attacks, Quantum channel tampering scenarios
6	Software Framework / Prototype	End-to-end implementable system	Simulation environment, Verification interface, Threat detection dashboard, Logging of security events

Organization	Egreen Quanta LLP
Category	Software
Theme	Transportation (1), Sustainability (2), Health (3), Education (4), Security (5)
Dataset	Public/Open
Video	NA
Title	1. Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization 2. Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization 3. Hybrid Quantum Machine Learning Platform for Early Disease Detection 4. AI-Based Interactive Quantum Algorithm Learning Platform 5. Quantum-Inspired Cyber Threat Detection for Digital Signature Security
Deliverables	Prototype, Code, Documentation